



TECHNICAL MAINTENANCE MANUAL

Lift Maintenance Manual - UH-LIFT-104

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|----------------|-----------------------------------------------------|
| Document ID    | MAN-005                                             |
| Version        | v1.0.0                                              |
| Client         | UrbanHive Facility Technologies Pvt. Ltd.           |
| Site           | GreenPark Meridian Residency, Whitefield, Bengaluru |
| Effective date | 2024-07-01                                          |
| Review cycle   | Annual or after any major incident                  |

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## 1. Equipment Overview and Technical Specifications

Passenger traction lift for high-rise residential service at GreenPark Meridian Residency. Machine-room-less (MRL) configuration with permanent magnet synchronous motor, VVVF drive, automatic rescue device (ARD), and EN 81-20 / IS 14665 compliance.

Model code: UH-LIFT-104

Asset registry: Record serial number and installation tower in AMIS before first energization.

Electrical supply: 415V AC 3-phase 50Hz, 32A isolator, RCCB 30mA, emergency UPS for ARD 24h

Environmental limits: Ambient 0-45°C operational; store spare parts below 35°C and 60% RH.

Standards: UrbanHive engineering bulletin UH-ENG-AMIS-2024; applicable Indian Standards as cited per asset class.

Documentation package: This manual, wiring diagram sheet WD-UH-LIFT-104, commissioning checklist CC-005, and spare parts catalogue SP-UH-LIFT-104.

Technicians must verify nameplate data matches AMIS asset record UH-LIFT-104 before applying procedures in this document. Discrepancies must be logged as a data quality ticket to UrbanHive Data Engineering within one business day.

## 2. Installation Guidelines

Verify shaft plumb within 15mm over full travel. Install guide rails to OEM torque 180 Nm. Commission door interlock chain, overspeed governor, and buffer engagement before passenger service. Upload as-built drawings to UrbanHive AMIS asset registry.

### Pre-commissioning checks:

- Verify incoming power quality: voltage balance within 2%, THD < 5% on dedicated feeder.
- Confirm earthing resistance  $\leq 1$  ohm at equipment location; bond to society main earth bar.
- Inspect shipping damage; photograph defects for vendor warranty within 48 hours.
- Complete mechanical alignment, torque logs, and insulation resistance tests per UrbanHive QC form UH-QC-INST-02.

Post-installation: Upload signed commissioning certificate, as-built photos, and warranty start date to AMIS. Tag physical asset with QR code linking to this manual revision v1.0.0.

## 3. Preventive Maintenance Schedule

UrbanHive mandates the following intervals for GreenPark Meridian. Adjust only with written approval from Head of Maintenance.

- Daily: lobby call panel test, alarm log review, cab lighting check.
- Weekly: door operator belt tension, sill gap measurement, pit sump pump test.
- Monthly: rope tension equalization, brake pad inspection, ARD battery load test.
- Quarterly: governor rope diameter check, guide shoe wear, safety gear functional test.
- Annual: statutory inspection with third-party certifier; full load test at 125% capacity.

Documentation: Each PM visit must generate a service log in AMIS with parts consumed, meter readings, and pass/fail checklist. Missed PM windows trigger automatic escalation to Facility Manager after 72 hours.

## 4. Troubleshooting Guide

Use isolation procedure: observe symptom ? confirm with meter/test ? apply remediation ? functional test ? document in AMIS.

If fault persists after two visits, escalate to UrbanHive vendor support with fault code and photos.

### Fault 01 - Code E101

Symptom: Car stops between floors with fault door zone

Remediation: Check door zone magnet alignment; verify encoder feedback; reset drive fault after confirming no obstruction in door track.

### Fault 02 - Code E204

Symptom: Repeated door reopening cycles

Remediation: Inspect door operator belt for glazing; adjust closing torque to 110 Nm; clean sill track and verify safety edge sensitivity at 30N.

### Fault 03 - Code E318

Symptom: ARD activation during mains failure

Remediation: Measure ARD battery terminal voltage (min 24V); replace cells if internal resistance > 45 mOhm; verify charging circuit diode pack.

### Fault 04 - Code E402

Symptom: Cab vibration above 2.5 m/s<sup>2</sup>

Remediation: Inspect guide shoe liner thickness (replace if < 3mm); check rail joint steps; balance car load distribution.

### Fault 05 - Code E515

Symptom: Leveling error > 10mm at floors

Remediation: Calibrate laser leveling sensor; verify rope elongation; adjust drive parameter F17 floor offset.

### Fault 06 - Code E607

Symptom: Overloads sensor nuisance trips

Remediation: Confirm rated load plate matches 8-person / 630 kg config; test load cell zero drift; inspect rope hitch equalization.

### Fault 07 - Code E708

Symptom: Controller communication timeout

Remediation: Reseat CAN bus terminators; inspect trunking for moisture; update firmware build UH-LIFT-CTRL-3.2.1.

### Fault 08 - Code E812

Symptom: Emergency stop loop open

Remediation: Trace E-stop chain through pit, car top, and machine room; replace fractured contact in COP station.

### Fault 09 - Code E903

Symptom: Motor thermistor alarm

Remediation: Measure winding insulation > 1 M $\Omega$ ; verify cooling fan amp draw; check drive heatsink airflow in MRL cabinet.

Fault 10 - Code E1102

Symptom: Unintended car movement with doors open  
Remediation: DO NOT return to service. Lock out drive contactor. Inspect door lock hook engagement and DL1/DL2 relay logic.

Fault 11 - Code E1204

Symptom: Rope slip detected on sheave  
Remediation: Check rope tension balance within 5%; inspect sheave groove wear profile; replace rope if crown diameter loss > 8%.

Fault 12 - Code E1301

Symptom: Fire recall mode not engaging  
Remediation: Test smoke lobby detector input to lift controller; verify fireman switch key operation; confirm recall wiring to BMS panel.

5. Spare Parts List

Maintain minimum stock at GreenPark Meridian central store (Basement B1, Room ST-04).

| Part No. | Description             | Min stock |
|----------|-------------------------|-----------|
| SP-L01   | Door operator belt kit  | 4         |
| SP-L02   | Guide shoe liner set    | 12        |
| SP-L03   | ARD battery 12V 7Ah     | 6         |
| SP-L04   | Door lock hook assembly | 2         |
| SP-L05   | Encoder disk module     | 2         |
| SP-L06   | Brake pad set           | 3         |

Reorder when stock falls below minimum. Use UrbanHive procurement portal PR-UH-OPS for approved vendors only.

6. Safety Warnings and Compliance

Only trained lift mechanics may access car top or pit. Lock-out/tag-out drive isolator before pit entry. Full-body harness mandatory on car top. Never bypass door interlocks. Comply with Karnataka Lift Act and NBC 2016. UrbanHive incident reporting within 30 minutes for any trapped passenger event.

Permit requirements: Hot work, confined space, work-at-height, and electrical isolation require digital permit in AMIS before start.

PPE minimum: Safety shoes IS 15298, helmet IS 2925, eye protection IS 5983, gloves appropriate to task.

Incident reporting: Any injury, near-miss, or property damage - complete IR-01 within 30 minutes; notify security desk ext. 9911.

7. Manufacturer Contact and UrbanHive Support

OEM technical support: +91-80-5555-0005 (08:00-20:00 IST)

Email: support-uh-lift-104@urbanhive-vendor.example.com

UrbanHive Ops escalation: maintenance@urbanhiveops.example.com - Priority 1 response within 4 business hours

AMIS helpdesk: Log ticket category `Lift` with asset\_id from tower registry

Emergency (life safety assets): 24x7 hotline +91-80-5555-0911

Revision history: v1.0.0 - Initial release for GreenPark Meridian AMIS integration and UrbanHive contractor certification programme.

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*End of document MAN-005*